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Udacity MSc Capstone – Data Workflow

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Overview

This project conducts a complete statistical analysis using a combined public cybersecurity mapping dataset from Data.gov/NIST. I combined the AMI, DER, and DGM CSV resources into a single analysis table and performed descriptive statistics, visualization, and hypothesis testing to evaluate cross-domain pattern differences. The goal was to build a reproducible statistical baseline that can support later machine learning and decision-support work.

Dataset: Mapping of NIST Cybersecurity Framework Subcategories to NESCOR Threat Scenarios (AMI/DER/DGM)

Source link: https://catalog.data.gov/dataset/mapping-of-nist-cybersecurity-framework-subcategories-to-threat-scenarios-from-the-nationa-e8b41

Project repository: https://github.com/Ohara124c41/statistical-analysis-NIST-CFS

Dataset Description

The dataset represents mappings between NESCOR electric-sector threat scenarios and NIST CSF subcategories across three domains: AMI, DER, and DGM. I used a direct vertical concatenation of the three public CSV files and added provenance fields so each row remains traceable to its source domain and file. The resulting analysis file has 502 rows and 15 columns, satisfying the project requirements (>=500 rows, >=6 columns, numeric and categorical variables present). Key analysis variables were scenario_id (numeric), domain (categorical), csf_subcategory (categorical), and nist_function (categorical function prefix derived from CSF subcategories).

Methods

I followed an initial data analysis workflow by first checking structure, variable types, missingness patterns, and category frequencies before inferential testing, consistent with reproducible IDA recommendations (Lusa et al., 2024; Baillie et al., 2022). The implementation stack is Python in Jupyter Notebook, with NumPy/Pandas for data handling, SciPy for inferential statistics, and Matplotlib/Seaborn for visualization (Python Software Foundation, n.d.; Project Jupyter, n.d.; Harris et al., 2020; McKinney, 2010; Virtanen et al., 2020; Hunter, 2007; Waskom, 2021). Descriptive statistics included numeric summaries, categorical counts, top-category frequency tables, and domain balance. Visualizations were chosen to answer distinct questions: Figure 1 shows domain composition, Figure 2 shows numeric variable behavior for scenario_id, Figure 3 shows domain-by-function frequency structure, Figure 4 compares text-length distributions for mitigation descriptions, and Figure 5 evaluates long-tail concentration in subcategory frequencies.

For inference, I used a chi-square test of independence because the core question is whether two categorical variables (domain and nist_function) are associated. The null hypothesis was that domain and nist_function are independent; the alternative was that they are associated. I also reported Cramer's V as an effect-size estimate and added robustness diagnostics: expected-cell checks, standardized residuals, bootstrap confidence interval for effect size, and Holm-corrected pairwise domain contrasts.

Test Selection Justification

The primary research question is: "Does NIST CSF function distribution vary by domain (AMI, DER, DGM)?" Both variables in this question (domain, nist_function) are nominal categorical variables, so the natural baseline test is a chi-square test of independence on a contingency table (Agresti, 2013; SciPy Developers, n.d.-a). This test directly evaluates whether observed cross-tabulated counts differ from independence expectations and therefore matches the analytical question.

I reported Cramer's V to quantify association magnitude rather than relying only on statistical significance (Agresti, 2013; SciPy Developers, n.d.-b). I also used standardized residuals to identify which table cells drive the omnibus chi-square signal (Agresti, 2013). Because multiple pairwise domain comparisons were examined, Holm correction was applied to control family-wise error (Holm, 1979). To assess uncertainty of effect size under weaker parametric assumptions, I added a bootstrap confidence interval for Cramer's V (Efron & Tibshirani, 1993; SciPy Developers, n.d.-c). For text-length comparison across domains, Kruskal-Wallis and pairwise Mann-Whitney tests were used as nonparametric complements (Kruskal & Wallis, 1952; SciPy Developers, n.d.-d; SciPy Developers, n.d.-e).

Results

The ingestion and validation checks passed project constraints (rows_at_least_500=True, columns_at_least_6=True, numeric/categorical variables present, domains_present=True). Domain counts were AMI=191, DGM=166, DER=145. NIST function distribution was highly concentrated in PR (354 rows, 70.5%), followed by DE (69), missing/unmapped (60), ID (18), and RC (1).

The chi-square test found evidence of association between domain and nist_function (chi-square=18.1818, df=6, p=0.005793), so independence was rejected at alpha=0.05. Effect size was modest (Cramer's V=0.1434), and bootstrap uncertainty remained above zero (95% CI: 0.1136 to 0.1981), indicating a stable but not large association. Pairwise Holm-adjusted contrasts showed significant differences for AMI vs DER (p_holm=0.001884) and DER vs DGM (p_holm=0.019172), but not for AMI vs DGM (p_holm=0.362546).

Figure 1 shows domain representation is reasonably close but not perfectly balanced. Figure 3 shows PR dominates all domains with visibly different DE/ID proportions, which aligns with the significant chi-square result. Figure 4 shows similar ECDF trajectories for mitigation-text length across domains, matching the non-significant nonparametric comparison. Figure 5 shows a long-tail subcategory structure where a few labels occur frequently and many labels are sparse.

Assumption diagnostics showed sparse expected counts in the contingency table (33.3% of cells expected <5 and 25.0% expected <1), so the asymptotic p-value should be interpreted cautiously and together with effect size and resampling diagnostics (Agresti, 2013). Text-length analysis of potential_mitigations found no domain-level distribution difference (Kruskal-Wallis H=0.4313, p=0.806018; all Holm-corrected pairwise Mann-Whitney tests non-significant).

Baseline vs Experimental Comparison (Design and Performance Evaluation)

Baseline (Section 5) used a single omnibus chi-square test with effect size:

- chi-square=18.1818, df=6, p=0.005793

- Cramer's V=0.1434

- Decision: reject H0 of independence

Experimental enhancement (Section 5B) added robustness and localization:

- Assumption diagnostics: 33.3% expected cells <5; 25.0% <1

- Bootstrap CI for Cramer's V: [0.1136, 0.1981]

- Holm-corrected pairwise contrasts: AMI vs DER significant; DER vs DGM significant; AMI vs DGM not significant

- Residual localization: DER-related function mix differences contribute strongly to the global signal

- Nonparametric text-length check: no domain-level difference (p=0.806018)

What changed and why this comparison is meaningful:

The baseline test answered only a binary question ("is there any association?"). The experimental layer showed where association is concentrated (DER contrasts), how large it is (modest), and how sensitive inference is to sparse expected counts. This comparison matters operationally because it converts a yes/no significance statement into actionable, domain-specific risk insight while making uncertainty explicit.

Interpretation for a Non-Technical Audience

The main takeaway is that the three infrastructure domains do not map to cybersecurity functions in exactly the same way, but the difference is moderate rather than extreme. In practice, this means governance and control coverage may vary by domain and should not be assumed identical across AMI, DER, and DGM. At the same time, not every observed difference is large enough to imply major practical separation, so decisions should combine statistical significance with effect-size context. The text-based mitigation descriptions appear similarly distributed in length across domains, suggesting that differences are more about control-function mapping than writing-length style.

Limitations and Potential Bias

A primary limitation is that this is a curated mapping dataset, not operational event logs, so conclusions describe mapping structure rather than real-world incident rates. A second limitation is category sparsity and long-tail behavior (many rare subcategories), which can destabilize fine-grained inference. A key potential bias is treating unmatched or missing function labels as if they were random missingness; if missingness differs by domain, association estimates can be distorted. Another risk is over-emphasizing p-values without practical context; I reduced this by reporting effect sizes, confidence intervals, and corrected pairwise tests.

Ethical and Risk Implications for Critical Infrastructure

Because this analysis concerns electric-sector cybersecurity mappings, statistical misinterpretation can create real downstream harm. If domain-specific differences are overstated, governance teams may misallocate scarce security resources toward the wrong domain; if differences are understated, important gaps may remain under-prioritized. Sparse-table assumptions also create a false-confidence risk: a significant p-value without assumption checks could be read as stronger evidence than warranted. Finally, mappings themselves encode expert judgment, so any embedded subjectivity can propagate into policy, audit, and control-prioritization decisions if not explicitly acknowledged.

Reproducibility

TThe workflow is reproducible through a single notebook run (analysis.ipynb) with pinned dependencies in requirements.txt and deterministic combination logic for AMI/DER/DGM inputs. The combined CSV is generated in-project and can be rebuilt from source files in data/raw/ when needed.

Technology and Algorithm Citation Coverage

- Technologies referenced and cited: Python, Jupyter Notebook, NumPy, Pandas, SciPy, Matplotlib, Seaborn.

- Algorithms referenced and cited: chi-square independence test, Cramer's V, standardized residual diagnostics, bootstrap confidence intervals, Holm multiple-comparison correction, Kruskal-Wallis test, Mann-Whitney U test.

- Link references: URLs in References were checked on February 18, 2026.

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